

Modify the Firmware Parameters of the Car's Control Board

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1. Hardware Connection
2. Modify Parameters
3. Write Configuration

This course requires that the car has already been flashed with the factory firmware before you can configure the firmware parameters. The purpose of modifying the control board's firmware parameters is to make the car's functions better suit individual needs. The factory firmware comes with default parameters, which do not need to be modified unless necessary.

The car's control board has been flashed with the factory firmware at the factory. If you have flashed other firmware, please re-flash the factory firmware according to the course [13.Control Board Course-1.Development environment setup-4.Burning STM32 firmware using serial port]. The factory firmware is located in the [Firmware_DriverFile\Control Board Factory Firmware] folder.

This course uses the Jetson Orin series mainboard as an example; the operation is the same for other mainboards.

Note: Before running the configuration modification script, you need to close the microros agent first.

1. Hardware Connection

Ensure that the Type-C USB Connect interface of the control board is connected to the USB port of the mainboard.

2. Modify Parameters

Open the system terminal, find and open the [config_robot.py] file in the user directory.

```
vim ~/config_robot.py
```

Scroll to the bottom of the file. The configuration parameters mainly include set_ros_domain_id, set_ros_namespace, set_motor_pid_parm, set_imu_yaw_pid_parm, set_ros_scale_line, set_ros_scale_angular, and set_arm_mid_value.

If you need to modify a parameter, please remove the comment symbol in front of the corresponding function.

```
# robot.set_ros_domain_id(30)
# robot.set_ros_namespace("")
# robot.set_motor_pid_parm(0.8, 0.06, 0.5)
# robot.set_imu_yaw_pid_parm(0.6, 0, 0.3)

# robot.set_ros_scale_line(1.0)
# robot.set_ros_scale_angular(1.0)

# arm_mid_value = [2000, 2000, 2000, 2000, 1486, 3100]
# robot.set_arm_mid_value(arm_mid_value)
```

`set_ros_domain_id` sets the car's ROS DOMAIN ID, with a range of 0~100. If there are multiple devices on the local network, set a different ROS DOMAIN ID for each to avoid interference. Note that it must be consistent with the ROS_DOMAIN_ID value in the system terminal's .bashrc file for communication.

`set_ros_namespace` sets the car's ROS namespace, mainly used for multi-car control functions on a local network.

`set_motor_pid_parm` sets the PID parameters for the car's motor speed.

`set_imu_yaw_pid_parm` sets the PID parameters for the car's direction calibration using the IMU.

`set_ros_scale_line` sets the scaling ratio for the car's ROS linear velocity.

`set_ros_scale_angular` sets the scaling ratio for the car's ROS angular velocity.

`set_arm_mid_value` sets the median deviation of the robotic arm, mainly used to quickly restore the robotic arm's calibration values. It generally does not need to be modified. Note: If you modify this value, it will affect the calibrated values of the robotic arm. If you need to calibrate the robotic arm, please refer to the robotic arm calibration tutorial.

Here is an example of modifying the domain id to 31:

```
robot.set_ros_domain_id(31)
# robot.set_ros_namespace("")
# robot.set_motor_pid_parm(0.8, 0.06, 0.5)
# robot.set_imu_yaw_pid_parm(0.6, 0, 0.3)

# robot.set_ros_scale_line(1.0)
# robot.set_ros_scale_angular(1.0)

# arm_mid_value = [2000, 2000, 2000, 2000, 1486, 3100]
# robot.set_arm_mid_value(arm_mid_value)

# Please remove the comments. If you need to save the configuration.
robot.update_config_data()
```

3. Write Configuration

Note: Before running the configuration modification script, you need to close the microros agent first.

Open the system terminal and run the following command to start writing the configuration:

```
python3 ~/config_robot.py
```

```
Read device: Linux
Waiting to read the car type
car_type: 7
version: 1.1.2
domain_id: 31
ros_namespace:
ros_scale_line: 1.000
ros_scale_angular: 1.000
motor pid parm: 0.800, 0.060, 0.500
imu pid parm: 0.600, 0.000, 0.300
arm_mid: 2025,2077,1957,2017,1419,3005
```

You can see all the parameter values that are finally read.